STAT 505 Assignment Submission [L5]

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1. *(adapted from J&W Exercise 1.6)* Various measurements on air pollution in Los Angeles are recorded in the data set "air.csv". Variables are X_1 = wind, X_2 = solar radiation, X_3 = CO, X_4 = NO, X_5 = NO_2, X_6 = O_3, and X_7 = HC.

```
data air;
infile "/air.csv" delimiter="," firstobs=2;
input x1-x7;
run;
```

(a) Explain briefly what X_{ij} and μ_j represent in terms of the variables here. Similarly, explain briefly what σ_{jk} represents. What does it mean if $\sigma_{jk} = 0$?

X_{ij} represents the ith observation for the jth variable. Similarly, μ_j represents the mean for the jth variable. σ_{jk} represents the covariance between variables j and k. If $\sigma_{jk} = 0$ then that means there is zero covariance and that j and k are independent if they are normal random variables.

(b) Considering only CO, NO_2, O_3, and HC, can these variables be assumed to be multivariate normal? Justify your answer with appropriate plots.

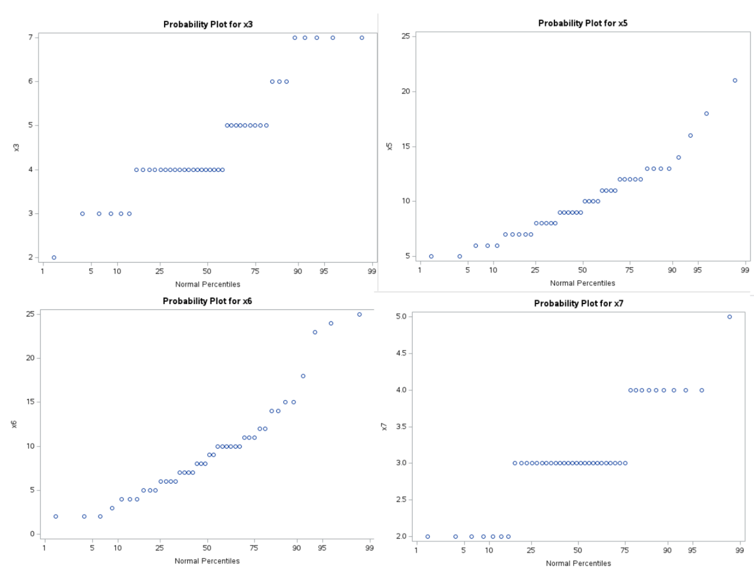

Figure 1: SAS solution for part B.

Based on the normal probability plots seen in the Figure 1, assumptions for normal probability are broken for CO (x3) and HC (x7), as evidenced by the deviation from the diagonal of the plot.

(c) Provide a matrix of scatterplots for the four variables above in (b). Also, report the numeric correlation for each pair of variables. Is any pair significantly correlated? Answer this with separate confidence intervals of correlation. Use a Bonferroni adjustment for multiplicity so that your confidence for all intervals simultaneously is 95%.

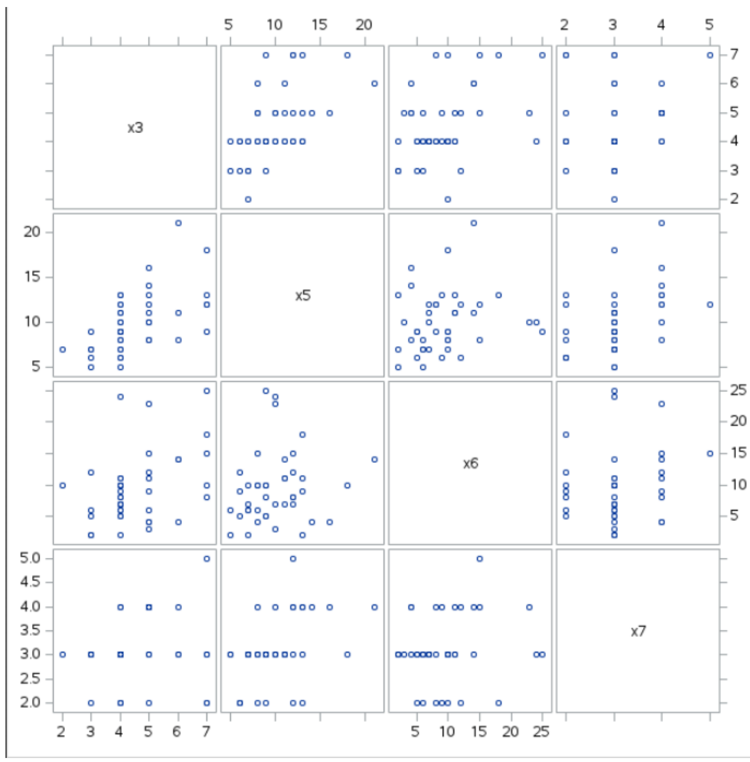

Figure 2: SAS solution for part C.

pair	r	lower_r	upper_r	Significance
X3 vs X5	0.55658	0.18077	0.79057	Significantly Correlated
X3 vs X6	0.41093	-0.00836	0.70733	Not Significant
X3 vs X7	0.16603	-0.27059	0.54600	Not Significant
X5 vs X6	0.16664	-0.27001	0.54644	Not Significant
X5 vs X7	0.44777	0.03680	0.72919	Significantly Correlated
X6 vs X7	0.15445	-0.28157	0.53761	Not Significant

Figure 3: SAS solution for part C with Bonferroni adjustment.

As seen in Figures 2 and 3, only the variables that are correlated are the pair CO (x3) and NO (x5) and the pair NO (x5) and HC (x7).

2. *(adapted from J&W Exercise 5.7)* The variables in "sweat.csv" are sweat rate, sodium, and potassium for a sample of $n = 20$ females.

```
data sweat;
infile "/sweat.csv" delimiter="," firstobs=2;
input rate na k;
run;
```

(a) Find 90% individual one-at-a-time confidence intervals for the population means of the three variables using the univariate t-distribution.

variable	xbar	lower_90	upper_90
k	9.965	9.2286	10.7014
na	45.400	39.9349	50.8651
rate	4.640	3.9839	5.2961

Figure 4: SAS solution for part A.

As depicted in Figure 4, the individual confidence intervals are (9.2286, 10.7014) for k, (39.9349, 50.8651) for na, and (3.9839, 5.2961) for rate.

(b) Find 90% "simultaneous" confidence intervals for the population means of the three variables (using the multiplier based on the F distribution).

variable	xbar	f_multiplier	lower_90_sim	upper_90_sim
k	9.965	2.85877	8.7475	11.1825
na	45.400	2.85877	36.3646	54.4354
rate	4.640	2.85877	3.5553	5.7247

Figure 5: SAS solution for part B.

As depicted in Figure 5, the simultaneous confidence intervals are (8.7475, 11.1825) for k, (36.3646, 54.4354) for na, and (3.5553, 5.7247) for rate.

(c) How do the interpretations of these two sets of intervals compare with one another? Why are those in (b) wider?

The multiplier in part (a) is 1.729 while the multiplier in part (b) is 2.859. The larger value in part (b) yields a larger confidence interval.

(d) The researchers were also interested in whether the population mean vector differed from $\mu_0 = [4, 50, 10]'$, and in a formal test they rejected this vector at the $\alpha = .10$ level. Is this consistent with your results in (b) above? Why or why not?

No, this is inconsistent with the findings from part (b) as the vector $[4, 50, 10]$ falls within the 90% confidence interval for all three variables.